

Digital Systems EE 2001 Solutions

Tutorial 1 March 2023

1. Convert the following numbers to the specified number systems. (three decimal points)
(Neethu)

a. $(28.8)_{10} = ()_2$

a) $(28.8)_{10} = ()_2$

for Integer part

	Quotient	remainder
$28/2$	14	0
$14/2$	7	0
$7/2$	3	1
$3/2$	1	1

Ans = 11100

$(28.8)_{10} = (11100.11001)_2$

for decimal part

	Ans	Integer part
0.8×2	1.6	1
0.6×2	1.2	1
0.2×2	0.4	0
0.4×2	0.8	0
0.8×2	1.6	1

Ans = 11001

- b. $(-30.5)_{10} = ()_2$ (represent in 2's complement form)

Negative numbers can be represented in 2's complement form. First we have to check how many bits are required to represent -30.5. If we have n bits we can have $-(2^{n-1})$ to $2^{n-1} - 1$.

Eg: if $n = 6$, we can represent -32 to 31

So in order to represent -30.5 we need 6 bits

Negative numbers should be represented in 2's complement form.

$$(-30.5)_{10} = ()_2$$

step 1: find the binary equivalent of 30.5

$$\begin{array}{r} 2 \overline{) 30} = 0 \\ 2 \overline{) 15} = 1 \\ 2 \overline{) 7} = 1 \\ 2 \overline{) 3} = 1 \\ 1 \end{array} \quad \begin{array}{l} 0.5 \times 2 = 1.0 \\ 0.0 \times 2 = 0.0 \end{array}$$

$$(30.5)_{10} = (11110.1)_2$$

$$(30)_2 \rightarrow 011110.1$$

$$\begin{array}{r} \text{Take its complement} \rightarrow 100001.0 \\ \hline 100001.1 \end{array}$$

Now let us cross check this

$$\begin{array}{ccccccc} 1 & 0 & 0 & 0 & 0 & 1 & . & 1 \\ 1 & 1 & 1 & 1 & 1 & \downarrow & \downarrow & \\ -(2^5) & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & 2^{-1} & \end{array}$$

$$\begin{aligned} & 1 \times -(2^5) + 1 \times 2^0 + 1 \times \frac{1}{2} \\ & -32 + 1 + 0.5 = \underline{\underline{-30.5}} \end{aligned}$$

a. $(1001.110)_2 = ()_{10}$

$$c) (1001.110)_2 = ()_{10}$$

$$= 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times \frac{1}{2} + 1 \times \frac{1}{2^2} + 0 \times \frac{1}{2^3}$$

$$= 8 + 1 + 0.5 + 0.25$$

$$= \underline{\underline{9.75}}$$

b. $(640.7)_8 = ()_{10}$

$$(640.7)_8 = ()_{10} \quad \begin{matrix} 6 & 4 & 0 & . & 7 \\ 8^2 & 8^1 & 8^0 & & 8^{-1} \end{matrix}$$

$$\begin{aligned} &= 6 \times 8^2 + 4 \times 8^1 + 0 \times 8^0 + 7 \times 8^{-1} \\ &= 384 + 32 + 0 + 0.875 \\ &= \underline{\underline{416.875}} \end{aligned}$$

c. $(0.528)_{10} = ()_8$

$$(0.528)_{10} = ()_8$$

$$0.528 \times 8 = 4.224 \quad 4$$

$$.224 \times 8 = 1.792 \quad 1$$

$$.792 \times 8 = 6.336 \quad 6$$

$$.336 \times 8 = 2.688 \quad 2$$

$$\text{Ans} = (0.4162)_8$$

d. 7562.45 to octal

(a) $(7562.45)_{10} \rightarrow ()_8$

$(7562)_{10} \rightarrow (a)_8$

Find $a \Rightarrow$

8	7562	
8	945	2
8	118	1
8	14	6
	1	6

$(7562)_{10} = (16612)_8$

$(0.45)_{10} = (b)_8$

Find $b = ?$

$\begin{array}{r} 0.4 \\ \times 8 \\ \hline 3.2 \end{array}$	$\begin{array}{r} 0.2 \\ \times 8 \\ \hline 1.6 \end{array}$	$\begin{array}{r} 0.6 \\ \times 8 \\ \hline 4.8 \end{array}$	$\begin{array}{r} 0.8 \\ \times 8 \\ \hline 6.4 \end{array}$	$\begin{array}{r} 0.60 \\ \times 8 \\ \hline 4.8 \end{array}$	$\begin{array}{r} 0.8 \\ \times 8 \\ \hline 6.4 \end{array}$
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$(0.45)_{10} = (0.346314631)$

$(7562.45)_{10} = (16612.34631)$

$a = (16612)_8$

e. 1938.257 to hexadecimal

$$(b) (1938.257)_{10} = (\quad)_{16}$$

$$(1938)_{10} = (\quad)_{16}$$

16	1938	
16	121	2
	7	9

$$\text{C} = 792$$

$$(1938)_{10} = (792)_{16}$$

$$(0.257)_{10} = (d)_{16}$$

Find d = ?

$$\begin{array}{r} 0.257 \\ \times 16 \\ \hline 4.112 \end{array}$$

$$\begin{array}{r} 0.112 \\ \times 16 \\ \hline 1.792 \end{array}$$

$$\begin{array}{r} 0.792 \\ \times 16 \\ \hline 12.672 \end{array}$$

$$\begin{array}{r} 0.672 \\ \times 16 \\ \hline 10.752 \end{array}$$

$$\begin{array}{r} 0.752 \\ \times 16 \\ \hline 12.032 \end{array}$$

$$\begin{array}{r} 0.032 \\ \times 16 \\ \hline 0.512 \end{array}$$

$$\begin{array}{r} 0.512 \\ \times 16 \\ \hline 8.192 \end{array}$$

$$\begin{array}{r} 0.192 \\ \times 16 \\ \hline 3.072 \end{array}$$

$$\begin{array}{r} 0.072 \\ \times 16 \\ \hline 1.152 \end{array}$$

$$(0.41CAC0831 \dots)$$

$$(1938.257)_{10} = (792.41CAC0831 \dots)_{16}$$

f. 175.175 to binary.

$$(c) (175.175)_{10} = (\quad)_2$$

$$(175)_{10} = (\quad)_2$$

2	175	
2	87	1
2	43	1
2	21	1
2	10	1
2	5	0
2	2	1
	1	0

$$(10101111)_2 = (175)_{10}$$

$$(0.175)_{10} = (\quad)_2$$

$$\begin{array}{r} 0.175 \\ \times 2 \\ \hline 0.35 \end{array}$$

$$\begin{array}{r} 0.35 \\ \times 2 \\ \hline 0.70 \end{array}$$

$$\begin{array}{r} 0.7 \\ \times 2 \\ \hline 1.4 \end{array}$$

$$\begin{array}{r} 0.4 \\ \times 2 \\ \hline 0.8 \end{array}$$

$$\begin{array}{r} 0.8 \\ \times 2 \\ \hline 1.6 \end{array}$$

$$\begin{array}{r} 0.6 \\ \times 2 \\ \hline 1.2 \end{array}$$

$$\begin{array}{r} 0.2 \\ \times 2 \\ \hline 0.4 \end{array}$$

$$\begin{array}{r} 0.4 \\ \times 2 \\ \hline 0.8 \end{array}$$

$$\begin{array}{r} 0.8 \\ \times 2 \\ \hline 1.6 \end{array}$$

$$(0.175)_{10} = (001011001 \dots)_2$$

$$(175.175)_{10} = (10101111.001011001 \dots)_2$$

2. Determine the value of x if $(211)_x = (152)_8$. (Bhupender)

$$(211)_x = (152)_8$$

For calculating the value of x convert both sides into decimal

$$2 \cdot x^2 + 1 \cdot x + 1 \cdot x^0 = 1 \cdot 8^2 + 5 \cdot 8 + 2 \cdot 8^0$$

$$2x^2 + x + 1 = 64 + 40 + 2$$

$$2x^2 + x + 1 = 106$$

$$2x^2 + x - 105 = 0$$

$$x = -7.5, 7$$

The Base can never be -ve so $x = 7$

3. The binary numbers listed have a sign in the leftmost position and if they are negative they are in 2's complement form. Perform the following arithmetic operations and represent your answers in binary and decimal form. (Neethu)

- $1010.011 + 1110.001$
- $001110 + 110010$
- $111.001 + 001.010$
- $101011 + 100110$

a) $1010.011 + 1110.001$

$$\begin{array}{r} 1010.011 + \\ 1110.001 \\ \hline 1000.100 \end{array}$$

① 1000.100
↓
carry bit

last 2 carry bits are the same, so there is no overflow.

$$\begin{array}{r} 1000.100 \rightarrow \\ 0111.011 + \\ \hline 0011.100 \end{array}$$

$$(0011.100)_2 \Rightarrow 7.5$$

so answer is -7.5

Since there is no overflow, we neglected the carry

Verification

$$1010.011$$

$$-(2^3) 2^2 2^1 2^0 \quad 2^{-1} 2^{-2} 2^{-3}$$

$$-(2^3) \times 1 + 2 \times 1 + 1 \times \frac{1}{2} + 1 \times \frac{1}{2^3}$$

$$-8 + 2 + 0.25 + 0.125$$

$$-6 + 0.375 = -5.625$$

$$-5.625 + -1.875 = -7.5$$

$$1110.001$$

$$(-2^3) 2^2 2^1 2^0 \quad 2^{-1} 2^{-2} 2^{-3}$$

$$-8 + 4 + 2 + 0.125$$

$$-2 + 0.125$$

$$-1.875$$

b) $001110 + 110010$

$$\begin{array}{r} 001110 \\ 110010 \\ \hline 1000000 \end{array}$$

last two carry bits are the same so discard the carry

Ans is $(000000)_2 = (0)_{10}$

c) $111.001 + 001.010$

$$\begin{array}{r} 111.001 \\ 001.010 \\ \hline 1000.011 \end{array}$$

Both carries are same and so neglect the carry and answer is a positive number and is represented in original form

Ans is $(000.011)_2 \rightarrow (0.325)_{10}$

Cross checking.

$$\begin{array}{r} 111.001 \rightarrow \text{neg no:} \\ \underline{2^2 \quad 2^1 \quad 2^0 \quad 2^{-1} \quad 2^{-2} \quad 2^{-3}} \\ -(2)22 \quad 222 \end{array}$$

$$\begin{aligned} & -4 + 2 \times 1 + 2 \times 1 + \frac{1}{8} \\ & = -4 + 2 + 1 + 0.125 \\ & = -1 + 0.125 \\ & = \underline{\underline{-0.875}} \end{aligned}$$

$$\begin{array}{r} 001.010 \rightarrow \text{positive no} \\ \underline{2^2 \quad 2^1 \quad 2^0 \quad 2^{-1} \quad 2^{-2} \quad 2^{-3}} \end{array}$$

$$\begin{aligned} & = 1 \times 2^0 + 1 \times \frac{1}{4} \\ & = \underline{\underline{1.25}} \end{aligned}$$

$$\begin{aligned} & \Rightarrow -0.875 + 1.25 \\ & = \underline{\underline{0.375}} \end{aligned}$$

d) $101011 + 100110$

$$\begin{array}{r} 01110 \\ 101011 \\ 100110 \\ \hline 1010001 \end{array}$$

last two carry bits are not same so an overflow is detected

Here the carry cannot be discarded hence the operation can be performed only when original numbers are represented using 7 bits. instead of 6 bits. Represent the original no.'s using 7 bits and perform the addition.

$$\begin{array}{r} 101011 \rightarrow 1101011 + \\ 100110 \rightarrow 1100110 \\ \hline 11010001 \end{array}$$

$$\begin{array}{r} 1010001 \xrightarrow[2's \text{ complement}]{1} 0101110 + \\ \hline (0101110)_2 \Rightarrow (47)_{10} \end{array}$$

Ans = $(-47)_d$ is decimal

Note: When representing the number using extra bits replicate the MSB

Cross verification

$$\begin{array}{r} 101011 \\ -(2^5) 2^4 2^3 2^2 2^1 2^0 \\ -32 + 8 + 2 + 1 \\ \hline -21 \end{array}$$

$$\begin{array}{r} 100110 \\ -(2^5) 2^4 2^3 2^2 2^1 2^0 \\ -32 + 2^2 + 2^1 \\ -32 + 4 + 2 = -26 \end{array}$$

$$\begin{array}{r} \text{Ans} = -21 + \\ -26 \\ \hline -47 \end{array}$$

4. Given the two binary numbers $X = 1010100$ and $Y = 1000011$, perform the subtraction operation (Bhupender)
- $X - Y$
 - $Y - X$ using 2's complement.

(i) $X = 1010100$ $Y = 1000011$

$X - Y = X + (-Y) = X + 2's \text{ Complement of } Y$

2's Complement of $Y = 1's \text{ Complement} + 1$

$$\begin{array}{r} 011100 \\ + 1 \\ \hline 2's \text{ Comp. of } Y = 011101 \end{array}$$

$X + 2's \text{ Complement of } Y =$

$$\begin{array}{r} 1010100 \\ + 011101 \\ \hline \text{Carry} \leftarrow 10010001 \end{array}$$

While carry is 1, Discard it or Directly put bar on carry

$\therefore \text{Ans} = \bar{0}0010001$
sign bit

you can also verify it using decimal (Ans = +17)

$$(ii) \quad y - x = y + 2's \text{ Comp. of } x$$

$$2's \text{ Comp of } x \quad + \quad \begin{array}{r} 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \\ 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \\ \hline 1 \ 1 \ 0 \ 1 \ 1 \ 1 \ 1 \end{array}$$

$$1101111 = -(17)_{10}$$

$$\begin{aligned} & -64 + 32 + 8 + 4 + 2 + 1 \\ & = -64 + 47 = -17 \end{aligned}$$

5. Perform the arithmetic operation $(+42) + (-13)$ and $(-42) - (-13)$ in binary using the signed 2's complement representation for negative numbers. (Bhupender)

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$$(i) \quad 42 + (-13) = 42 + 2's \text{ Complement of } 13$$

$$+13 = (01101)_2 \xrightarrow{2's \text{ Comp}} (10011)_2$$

$$+42 =$$

2	42	
2	21	
2	10	1
2	5	0
2	2	1
	1	0

$$(0101010)_2$$

$$42 + 2's \text{ Comp of } 13$$

$$\begin{array}{r} 0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0 \\ + \ 1 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \\ \hline \end{array}$$

$$\textcircled{1} \ 0 \ 0 \ 1 \ 1 \ 0 \ 1$$

Carry

as Carry is 1 to directly Discard it & rest will be the answer

$$\text{Ans} = 001101$$

as in 2's comp the bits are extended & written same as MSB

$$-42 - (-13) = -42 + 13$$

2's Complement of $+42 + 13$

$$+42 = 0101010$$

$$\begin{array}{r} \text{2's Complement } +42 \rightarrow \\ 1010110 \\ + 0001101 \\ \hline 1100011 \end{array}$$

$$\text{Ans} = 1100011$$

6. Reduce the following boolean expressions. (Bhupender)

(a) $A'C' + ABC + AC'$

(b) $A'B(D' + C'D) + B(A + A'CD)$

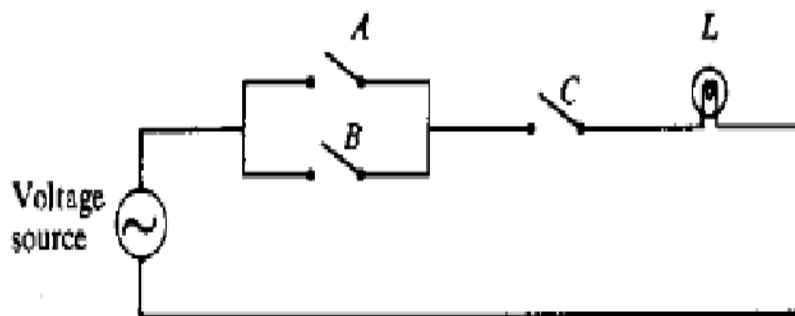
(c) $(A' + C)(A' + C')(A + B + C'D)$

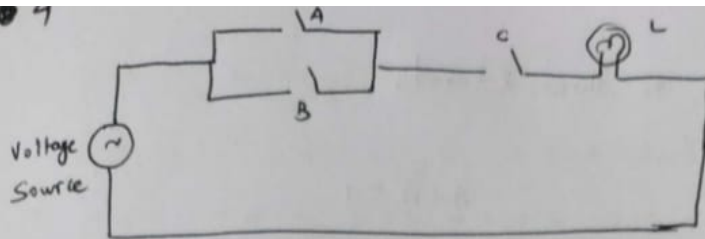
$$\begin{aligned} \text{a) } & A'C' + ABC + AC' \\ &= (A' + A)C' + ABC \\ &= C' + ABC \\ &= AB + C' \end{aligned}$$

$$\begin{aligned} \text{(b) } & A'B(D' + C'D) + B(A + A'CD) \\ &= A'BD' + A'BC'D + AB + A'BCD \\ &= A'BD' + AB + A'BD(C' + C) \\ &= A'B(D' + D) + AB \\ &= A'B + AB = \underline{B} \end{aligned}$$

$$\begin{aligned}
 (c) & (\bar{A} + C)(\bar{A} + \bar{C})(A + B + \bar{C}D) \\
 &= (\bar{A} \cdot \bar{A} + \bar{A} \cdot \bar{C} + C\bar{A} + C \cdot \bar{C})(A + B + \bar{C}D) \\
 &= (\bar{A} + \bar{A}\bar{C} + \bar{A}C)(A + B + \bar{C}D) \\
 &= \bar{A}(1 + \bar{C} + C)(A + B + \bar{C}D) \\
 &= (\bar{A})(A + B + \bar{C}D) \\
 &= \bar{A} \cdot A + \bar{A}B + \bar{A}\bar{C}D \\
 &= \bar{A}B + \bar{A}\bar{C}D \\
 &= \bar{A}(B + \bar{C}D)
 \end{aligned}$$

7. Express using binary logic the outcome of lighting up the bulb in the electrical switching circuit shown in the figure below. (Bhupender)





We have to Express the Binary logic notation

A	B	C	L
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

- when Both A and B are 0 then the ckt will be open
 or if C is 0 the ckt will be open

- when any one of A + B is 1 + C is 1 $L = 1$
 or both

Now write L in the form of ABC

~~$\bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$~~

we have to write the boolean form of L when it's 1

$$L = \bar{A}BC + A\bar{B}C + ABC \rightarrow \textcircled{1}$$

$$L = \bar{A}BC + ABC + A\bar{B}C$$

$$L = (\bar{A} + A)BC + A\bar{B}C$$

$$L = BC + A\bar{B}C$$

$$L = C(B + A\bar{B})$$

$$L = C(A + B) \rightarrow \textcircled{2}$$

$$A + \bar{A} = 1$$

$$B + A\bar{B} = (B + A)(B + \bar{B}) = B + AB$$

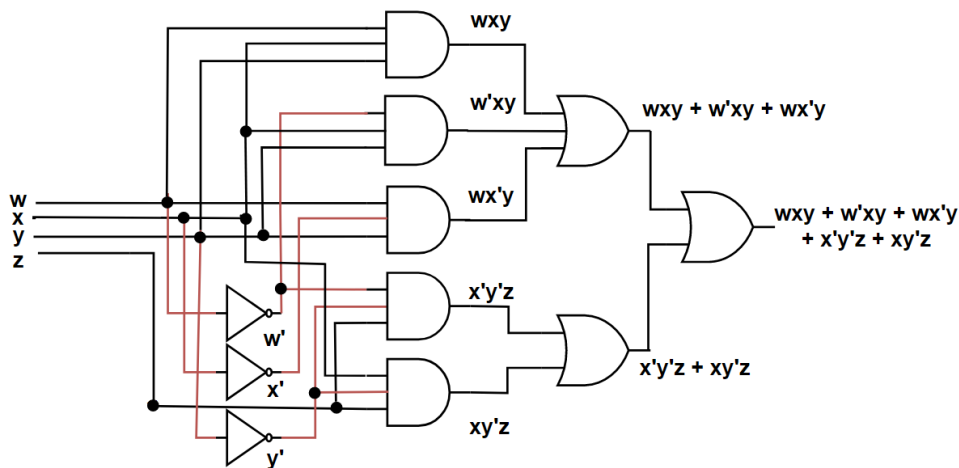
equation $\textcircled{1}$ is the answer to the question but eqⁿ $\textcircled{2}$ is the simplified form.

8. Given the following Boolean function (Neethu)

$$F = xy'z + x'y'z + w'xy + wx'y + wxy$$

- Obtain the truth table of the function
- Draw the logic diagram using the original boolean expression
- Simplify the function using to a minimum number of literals using Boolean Algebra
- Draw the logic diagram for simplified expression and compare the total number of gates with part a.

w	x	y	z	wxy	w'xy	wx'y	xy'z	x'y'z	o/p
0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	1	1
0	0	1	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0
0	1	0	1	0	0	0	1	0	1
0	1	1	0	0	0	1	0	0	1
0	1	1	1	0	0	1	0	0	1
1	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	1	1
1	0	1	0	0	1	0	0	0	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	0	0	0	0	0
1	1	0	1	0	0	0	1	0	1
1	1	1	0	1	0	0	0	1	1
1	1	1	1	1	0	0	0	0	1



$$F = xy'z + x'y'z + w'xy + wx'y + wxy$$

$$= y'z(x+x') + y(w'x + wx') + wxy$$

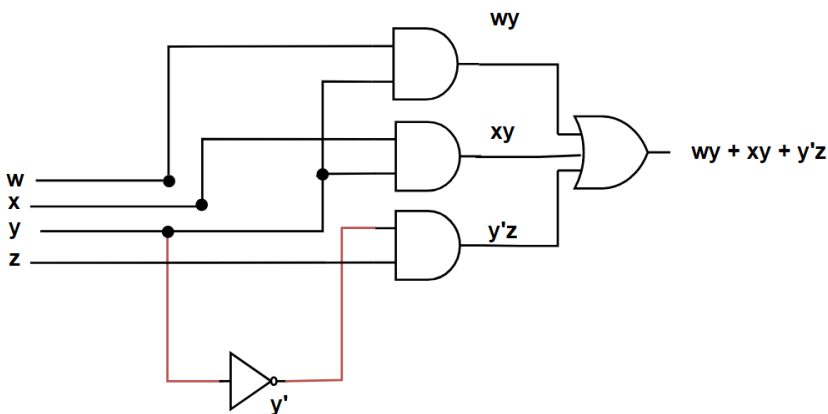
$$= y'z + y \overset{1}{(w'x + wx' + wx)}$$

$$= y'z + y(x(w'+w) + wx')$$

$$= y'z + y(x + x'w)$$

$$= y'z + y(x+w) = y'z + yx + wy$$

$$= \underline{wy + xy + y'z}$$



gates	Original	Simplified
3 input AND	4	0
2 input AND	0	3
3 input OR	1	0
2 input OR	2	1
NOT	4	1

9. Simplify the following expressions into given number of literals (Bhupender)

(a) $A'C' + ABC + AC'$ to three literals.

(b) $A'B(D' + C'D) + B(A + A'CD)$ to one literal.

(c) $(A' + C)(A' + C')(A + B + C'D)$ to four literals

a) $\bar{A}\bar{C} + ABC + A\bar{C}$ to three literals

$$\bar{A}\bar{C} + A\bar{C} + ABC$$

$$\bar{C}(A + \bar{A}) + ABC$$

$$\bar{C} + ABC$$

$$(\bar{C} + AB)(\bar{C} + C)$$

$$= \bar{C} + AB \quad (3 \text{ literals})$$

$$A + \bar{A} = 1$$

$$x + \bar{x}y = (x + \bar{x})(x + y)$$

$$C + \bar{C} = 1$$

b) $\bar{A}B(\bar{D} + \bar{C}D) + B(A + \bar{A}CD)$ into one literal

$$= \bar{A}B\bar{D} + \bar{A}B\bar{C}D + AB + \bar{A}BCD$$

$$= \bar{A}B(\bar{D} + CD) + \cancel{\bar{A}\bar{C}D}B \cancel{B\bar{A}CD} B(\bar{A}\bar{C}D + A)$$

$$= \bar{A}B[(\bar{D} + D)(\bar{D} + C)] + B((A + \bar{A})(A + \bar{C}D))$$

$$= \bar{A}B\bar{D} + \bar{A}BC + B(A + \bar{C}D)$$

$$= \bar{A}B\bar{D} + \bar{A}BC + AB + B\bar{C}D$$

$$B(\bar{A}\bar{D} + \bar{A}C + A + \bar{C}D)$$

$$A + \bar{A}C = (A + \bar{A})(A + C)$$

$$B(\bar{A}\bar{D} + A + C + \bar{C}D)$$

$$B(A + \bar{D} + C + D)$$

$$= B(A + C + 1)$$

$$1 + A = 1$$

$$= B(1) = B$$

$$\begin{aligned}
 (c) \quad & (\bar{A} + C)(\bar{A} + \bar{C})(A + B + \bar{C}D) \\
 &= (\bar{A} \cdot \bar{A} + \bar{A} \cdot \bar{C} + C \bar{A} + C \cdot \bar{C})(A + B + \bar{C}D) \\
 &= (\bar{A} + \bar{A} \bar{C} + \bar{A} C)(A + B + \bar{C}D) \\
 &= \bar{A}(1 + \bar{C} + C)(A + B + \bar{C}D) \\
 &= (\bar{A})(A + B + \bar{C}D) \\
 &= \bar{A} \cdot A + \bar{A} B + \bar{A} \bar{C} D \\
 &= \bar{A} B + \bar{A} \bar{C} D \\
 &= \bar{A}(B + \bar{C}D)
 \end{aligned}$$

10. Find the complement of the following expressions (Neethu)

- $xy' + x'y$
- $(AB' + C)D' + E$
- $AB(CD + C'D') + A'B'(C' + D)(C + D')$

a) $xy' + x'y$

$$\begin{aligned}
 (xy' + x'y)' &= (xy')' \cdot (x'y)' \\
 &= (x' + y)(x + y') \\
 &= x'x + x'y' + xy + yy' \\
 &= 0 + x'y' + xy + 0 \\
 &= \underline{\underline{xy + x'y'}}
 \end{aligned}$$

b) $(AB' + C)D' + E$

$$\begin{aligned}
 ((AB' + C)D' + E)' &= ((AB' + C)D')' \cdot E' \\
 &= [(AB' + C)' + (D')'] \cdot E' \\
 &= [(AB')' \cdot C' + D] \cdot E' \\
 &= ((A' + B) \cdot C' + D) E' = \underline{\underline{A'C'E' + BC'E' + DE'}}
 \end{aligned}$$

$$\begin{aligned}
& \Rightarrow \left[\underbrace{AB(CD + CD')} + \underbrace{A'B'(C'+D)(C+D')} \right]' \\
& = \left(AB(CD + CD') \right)' \cdot \left(A'B'(C'+D)(C+D') \right)' \\
& = \left[(AB)' + (CD + CD')' \right] \cdot \left[(A'B')' + (C'+D)' + (C+D')' \right] \\
& = \left[A' + B' + (CD' \cdot C'D) \right] \cdot \left[(A+B) + CD' + C'D \right] \\
& = \left[A' + B' + C'(D'D) \right] \cdot \left[A + B + CD' + C'D \right] \\
& = \underline{\underline{(A' + B') [A + B + CD' + C'D]}}
\end{aligned}$$

11. For the boolean function F given in the truth table, find the following (Bhupender)

- List the minterms of the function.
- List the minterms of F' .
- Express F in the sum of minterms in algebraic form.
- Simplify the function to an expression with a minimum number of literals.

x	y	z	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

a)

	x	y	z	F
0 →	0	0	0	0
1 →	0	0	1	0
2 →	0	1	0	1
3 →	0	1	1	1
4 →	1	0	0	0
5 →	1	0	1	0
6 →	1	1	0	1
7 →	1	1	1	1

When $F=0$ represents max terms
 $F=1$ represents min terms.

$$\text{min terms } F = \sum m(2, 3, 6, 7)$$

$$\text{max terms } F = \prod m(0, 1, 4, 5)$$

b) Min terms of $\bar{F}$ = max terms of F
 $\therefore \rightarrow \sum m(0, 1, 4, 5)$

c)

$$F = 2 + 3 + 6 + 7$$

$$= \bar{x}y\bar{z} + \bar{x}yz + x\bar{y}\bar{z} + xyz$$

d) Simplified form with minimum no. of literals

$$= \bar{x}y\bar{z} + \bar{x}yz + x\bar{y}\bar{z} + xyz$$

$$= \bar{x}y(\bar{z} + z) + xy(\bar{z} + z)$$

$$= \bar{x}y + xy$$

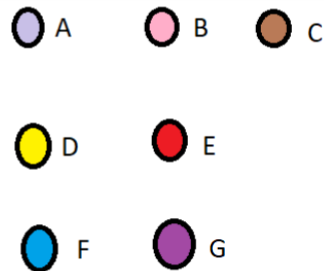
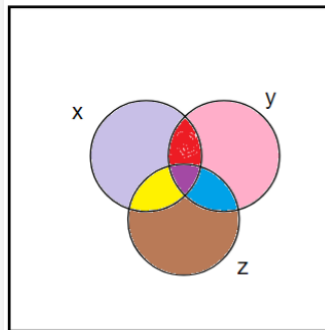
$$= (\bar{x} + x)y$$

$$= \underline{\underline{y}}$$

$(z + \bar{z}) = 1$

PRACTICE QUESTIONS

1. In the given Venn diagram, what does the coloured region (denoted by color codes A, B, C, D, E, F) represent? (Neethu)



- A) $x y' z'$
 B) $x' y z'$
 C) $x' y' z$
 D) $x z y'$
 E) $x y z'$
 F) $x' y z$
 G) $x y z$

2. What is the maximum value of A and minimum value of B in the below given expression $(2A7)_B = (1231)_5$ (Bhupender)

$$(2A7)_B = (1231)_5$$

we have to calculate minimum value of B and max. value of A.

$\therefore$ according to the concept of Bases

$$B > A$$

Convert both sides into Decimal equivalent

$$2 \cdot B^2 + A \cdot B + 7 = 1 \cdot 5^3 + 2 \cdot 5^2 + 3 \cdot 5 + 1$$

$$2B^2 + AB + 7 = 191$$

$$2B^2 + AB = 184$$

taking values of A + B which satisfy this eqⁿ

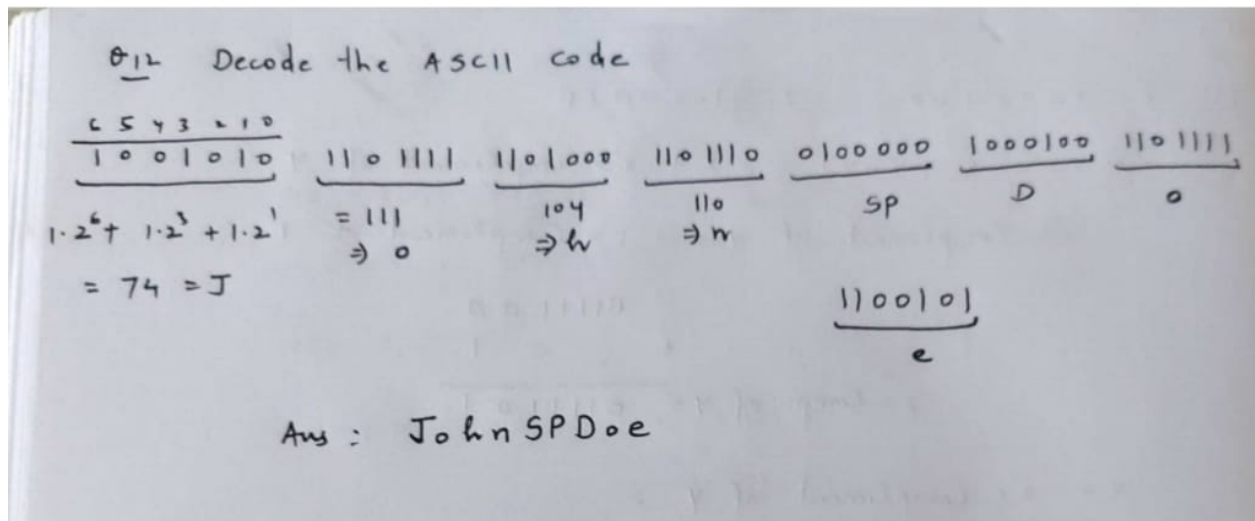
$$A = 7, B = 8$$

These values of A + B only satisfies the cond.

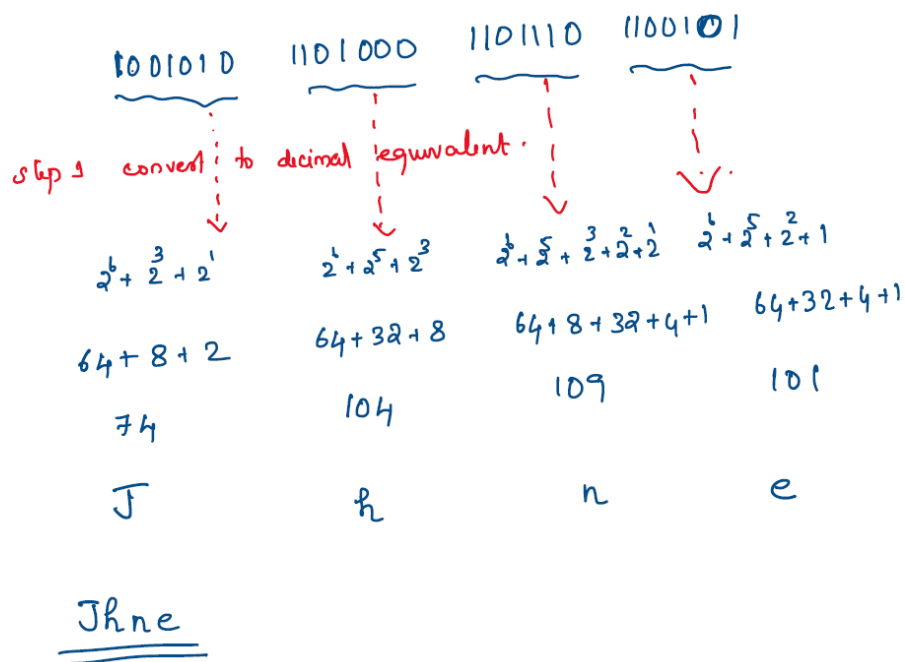
$$\underline{A < B} \text{ also satisfy the eqⁿ.}$$

3. Decode the following ASCII code: (Bhupender)

1001010 1101111 1101000 1101110 0100000 1000100 1101111 1100101



4. Decode the following ASCII code : 1001010 1101000 1101110 1100101(Neethu)



5. The state of a 12 bit register is 010110010110. What is its contents if it represents:(Neethu)

- Three decimal digits in BCD
- Three decimal digits in excess3 code
- Three decimal digits in 2421 code

9. 0101 1001 0110 group into groups of 4 bits

$\downarrow$ $\downarrow$ $\downarrow$
 5 9 6

In BCD it is (in decimal digits) 596

In excess 3 code

0101	1001	0110	
5	9	6	$\rightarrow$ BCD
3	3	3	convert to BCD
			subtract 3 to
2	6	3	find excess-3

0101 1001 0110 $\rightarrow$ rep 263 is excess-3.

In 2421 code

0101	1001	0110
<u> </u>	$1 \times 2 + 1 \times 1$	$4 + 2$
Not used	3	6

6. Express the following functions in sum of minterms(Neethu)

a. $F(x,y,z) = \prod(0, 3, 6, 7)$

b. $F(A,B,C,D) = \prod(0, 2, 6, 11, 13, 14)$

$$5) \quad a) P(m, y, z) = \prod (0, 3, 6, 7) \\ = \sum (1, 2, 4, 5) \quad \text{complement}$$

$$b) F(A, B, C, D) = \prod (0, 2, 6, 11, 13, 14) \\ = \sum (1, 3, 4, 5, 7, 8, 9, 10, 12, 15)$$

7. What is the largest binary number that can be obtained with 16 bits? What is its hexadecimal and decimal equivalent?(Neethu)

If we have 16 bits, the largest binary number that can be represented by 16 bits is

1111 1111 1111 1111

The decimal equivalent is $2^{16} - 1 = 65535$

In hexadecimal it is $(FFFF)_{16}$